Department of Electronics Engineering

IIT (BHU) Varanasi

Cost-Efficient Road Scene Reconstruction Using Deep Learning Techniques

Exploratory Project Report

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Abstract

Autonomous driving systems require precise, real-time understanding of the surrounding environment. Occupancy grid mapping is a foundational capability that allows a vehicle to distinguish between free space, occupied space, and unobserved regions without requiring class-specific object detectors. Radar sensors are increasingly attractive for this task due to their low cost, long range, and robustness under adverse weather conditions. However, the inherent sparsity and noise of clustered radar data make classical Bayesian filtering approaches unreliable. This project implements and evaluates a deep learning-based framework that formulates occupancy grid prediction from clustered radar data as a three-class semantic segmentation task (occupied, free, unobserved). The model uses a U-Net encoder-decoder architecture trained with the Lovász-Softmax loss to address class imbalance. Ground truth labels are generated automatically from lidar point clouds using ray tracing and morphological operations on the NuScenes real-world driving dataset. Temporal aggregation of up to 20 radar frames is incorporated to overcome data sparsity. Our implementation achieves a mean Intersection-over-Union (mIoU) of 0.4993, significantly outperforming classical inverse sensor model (ISM) baselines (mIoU ~0.19) and a ray-tracing baseline (mIoU 0.1744). These results confirm that data-driven occupancy grid learning from sparse radar is feasible and forms a cost-efficient alternative to LiDAR-centric perception pipelines.

1. Introduction

Safe autonomous vehicle navigation requires a continuous, accurate model of the surrounding environment. This environment model must correctly identify which areas of space are drivable (free), occupied by obstacles, or simply not observed by the sensor suite. This capability is often referred to as occupancy grid mapping, and it is essential for path planning, collision avoidance, and decision-making at all speeds.

Modern autonomous vehicles typically rely on a combination of cameras, LiDAR, and radar. LiDAR provides dense, accurate 3D point clouds and has become the primary sensor for occupancy grid mapping in research settings. However, LiDAR systems are expensive, fragile, and degrade significantly in adverse weather conditions such as heavy rain, fog, or snow. These limitations motivate the search for equally capable but more cost-efficient sensor modalities.

Radar is a compelling alternative: it is an order of magnitude cheaper than LiDAR, operates reliably over long ranges (>100 m), and maintains performance under fog, rain, and snow. Despite these practical advantages, radar produces inherently sparse and noisy measurements. Automotive radars typically operate at the clustered level — applying algorithms such as DBSCAN or CFAR to raw signals — yielding at most a few hundred detections per scan without any elevation information. This sparsity and noise make classical probabilistic methods brittle.

This project re-implements and extends the approach proposed by Sless et al. (ICCVW 2019), which is, to our knowledge, the first work to learn occupancy grid mapping directly from clustered radar data using deep learning. The key ideas are: (i) treat the 2D occupancy estimation as a semantic segmentation task in the Bird's-Eye-View (BEV) plane; (ii) aggregate multiple radar frames temporally to reduce the effective sparsity; and (iii) use lidar data

only at training time to generate ground truth labels, enabling radar-only inference at deployment. Our implementation is publicly available at https://github.com/0mM1shra/scene_rec.

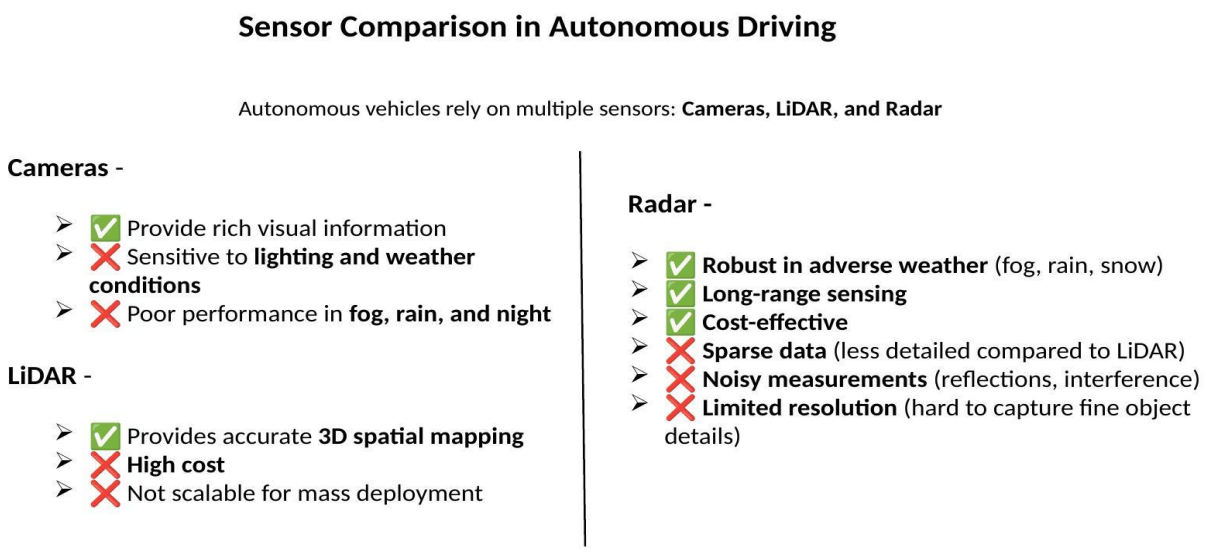

Figure 1. Sensor comparison for autonomous driving. Cameras offer rich visual information but fail in adverse weather. LiDAR provides 3D accuracy but is expensive. Radar is cost-effective and weather-robust but produces sparse, noisy measurements.

2. Related Work

2.1 Classical Occupancy Grid Mapping

The foundational framework for probabilistic occupancy grids was introduced by Elfes [1989] and later formalized with Bayesian filtering by Thrun et al. [2005]. In the classical setting, each grid cell maintains an independent occupancy probability updated by an Inverse Sensor Model (ISM) that maps raw sensor measurements to log-odds occupancy values. Delta-function and Gaussian ISM variants are commonly used for LiDAR and radar, respectively.

For radar specifically, the Gaussian ISM accounts for the angular and range uncertainty of radar detections. However, these hand-crafted functions require careful tuning and cannot easily adapt to the complex, scene-dependent noise characteristics of automotive radar. Prior works such as Werber et al. [2015] and Prophet et al. [2018] demonstrated occupancy grid generation from raw radar but relied exclusively on classical probabilistic methods without deep learning.

2.2 Deep Learning for Occupancy Mapping

Recent advances in convolutional neural networks have enabled data-driven occupancy grid estimation. Weston et al. [2018] proposed deep inverse sensor modelling from raw radar data but operated on a different data modality than clustered radar. Hoermann et al. [2018] presented a dynamic occupancy grid prediction system using

deep learning for LiDAR, demonstrating that learned approaches dramatically outperform classical filters. Lu et al. [2019] explored monocular camera-based semantic occupancy grids using variational encoder-decoder networks.

The work most closely related to ours is that of Sless et al. [2019], which first applied semantic segmentation architectures to clustered radar data. Their approach is distinctive in three respects: (i) it uses automatically generated lidar-derived ground truth rather than manual annotation; (ii) it explicitly models three occupancy states including the unobserved class; and (iii) it employs the Lovász-Softmax surrogate loss to directly optimize the mean Intersection-over-Union metric.

2.3 U-Net and Semantic Segmentation

The U-Net architecture [Ronneberger et al., 2015] introduced encoder-decoder networks with symmetric skip connections for dense per-pixel prediction. Originally designed for biomedical image segmentation, U-Net has since been widely adopted in autonomous driving tasks including BEV scene understanding. Its ability to preserve fine-grained spatial information via skip connections makes it well-suited for occupancy grid prediction, where precise boundary localization is critical.

3. Methodology

3.1 Problem Formulation

We formulate occupancy grid prediction as a dense semantic segmentation task in the Bird's-Eye-View (BEV) plane. The output grid y has spatial dimensions $H \times W = 215 \times 50$ with a cell resolution of $\alpha_x = \alpha_y = 40$ cm, covering a field of view from 0 to 86 m in range and −10 to +10 m in lateral extent from the ego vehicle. Each cell $y^{u,v}$ is assigned one of three labels:

- Occupied (label 1): The cell contains a static obstacle.
- Free (label 0): The cell is confirmed to be free of obstacles.
- Unobserved (label 2): The cell is beyond the sensor range or occluded; its state is unknown.

The network infers the posterior probability over these three states given k aggregated radar frames:

$$p(y \mid z_{t-k:t}, x_{t-k:t}, \theta)$$

where θ are the learnable network parameters, $z_{t-k:t}$ are the radar measurements from frames t−k to t, and $x_{t-k:t}$ are the corresponding ego-vehicle poses used for motion compensation.

Literature Survey

1. CenterFusion (2020)

- **Overview: Proposes a radar + camera fusion framework for 3D object detection using a center-based approach. It associates radar points with image detections and improves depth and velocity estimation.**

Advantages:

- **Improves detection accuracy (↑ NDS by ~12%)**
- **Better velocity estimation using radar Doppler**
- **Robust in adverse conditions (rain, fog)**

Disadvantages:

- **Still depends on camera (not radar-only)**
- **Complex sensor fusion pipeline**
- **Limited by radar sparsity and noise**

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2. BEV-GS / SLESS-type Road Reconstruction

- **Overview: Uses BEV (Bird's Eye View) representation and deep learning for road surface reconstruction, focusing on geometry + texture prediction for accurate mapping.**

Advantages:

- **High-accuracy road reconstruction (cm-level error)**
- **Real-time performance possible**
- **Good for occupancy grid / BEV tasks**

Disadvantages:

- **Often relies on camera/LiDAR inputs**
- **Not designed for radar-only input**
- **Struggles with sparse or noisy sensing modalities**

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3. GenRadar (2021)

- **Overview: A self-supervised model that learns to reconstruct camera-like scene understanding using only radar signals, exploiting cross-modal relationships.**

Advantages:

- **Works in low-visibility conditions**
- **Reduces dependency on cameras**

- **Learns cross-modal representations**

Disadvantages:

- **Generated outputs may lack fine visual details**
- **Training is complex (self-supervised learning)**
- **Still limited by radar information loss**

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4. RadarGen (2025)

- **Overview: Uses diffusion models to generate realistic radar point clouds from images, enabling data augmentation and simulation.**

Advantages:

- **Generates realistic radar data distributions**
- **Useful for training data augmentation**
- **Captures radar properties (RCS, Doppler)**

Disadvantages:

- **Requires camera input (not radar-only)**
- **Generative approach → not direct perception**
- **Computationally expensive**

Research Objective

- **Goal:** To replace expensive Lidar with Cost efficient Radar while maintaining same features.
- **Innovation:** Using deep learning methods to make our model predict ground truth by just Radar point cloud.

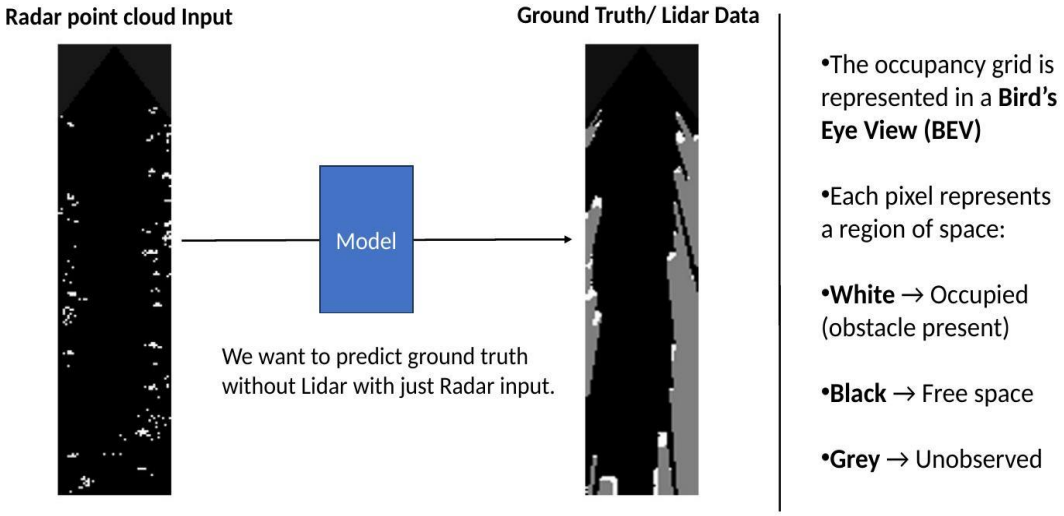

Figure 2. Research objective. Sparse radar point cloud input (left) is processed by the model to predict the ground truth occupancy grid (right), which captures free space (black), occupied regions (white), and unobserved areas (grey).

3.2 Data Preprocessing and Ground Truth Generation

Ground truth labels are generated from the NuScenes dataset using the following automated pipeline:

1. Lidar aggregation: All lidar frames in a scene are accumulated in global coordinates to form a dense point cloud, overcoming range limitations and temporary occlusions.
2. BEV projection: The aggregated cloud is projected onto the 2D H×W grid in the radar's coordinate frame. Low-density cells are removed via binary thresholding.
3. Morphological refinement: Dilation, hole filling, and erosion operations are applied to remove noise and smooth obstacle boundaries.
4. Ray tracing: For each radar viewpoint, rays are cast to assign free (cells before first return), occupied (cells at first return), or unobserved (cells after first return) labels.
5. Concave hull masking: A concave hull (alpha shape) is computed over the projected lidar cloud and used to mask regions where lidar coverage is absent. These cells receive an ignore label and are excluded from loss computation and evaluation.

Temporal aggregation is performed by warping k=20 consecutive radar frames to the current ego pose before stacking. Dynamic radar clusters are filtered using the velocity information provided by the NuScenes radars. Training examples are drawn from non-overlapping windows to minimize temporal correlation.

3.3 Network Architecture

The occupancy network (OccupancyNet) follows the U-Net encoder-decoder design with skip connections, implemented in PyTorch. The encoder progressively downsamples the input BEV radar image through three stages of double convolution blocks (each consisting of two 3×3 convolutions, batch normalization, and ReLU activations),

increasing the number of feature channels from 64 to 128 to 256 to 512. The decoder mirrors this with three upsampling stages that bilinearly interpolate feature maps and concatenate them with skip-connected encoder features before passing through another double convolution block. The output layer is a 1×1 convolution producing three-channel logits for the three occupancy classes.

The implementation from our repository (model.py) is reproduced below:

```
import torch, torch.nn as nn, torch.nn.functional as F

class DoubleConv(nn.Module):
    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.double_conv = nn.Sequential(
            nn.Conv2d(in_channels, out_channels, 3, padding=1),
            nn.BatchNorm2d(out_channels), nn.ReLU(inplace=True),
            nn.Conv2d(out_channels, out_channels, 3, padding=1),
            nn.BatchNorm2d(out_channels), nn.ReLU(inplace=True))
    def forward(self, x): return self.double_conv(x)

class RadarOccupancyNet(nn.Module):
    def __init__(self, n_channels=1, n_classes=3):
        super().__init__()
        self.inc   = DoubleConv(n_channels, 64)
        self.down1 = DoubleConv(64,  128)
        self.down2 = DoubleConv(128, 256)
        self.down3 = DoubleConv(256, 512)
        self.up1   = DoubleConv(512+256, 256)
        self.up2   = DoubleConv(256+128, 128)
        self.up3   = DoubleConv(128+64,  64)
        self.outc  = nn.Conv2d(64, n_classes, 1)

    def forward(self, x):
        x1 = self.inc(x)
        x2 = self.down1(F.max_pool2d(x1, 2))
        x3 = self.down2(F.max_pool2d(x2, 2))
        x4 = self.down3(F.max_pool2d(x3, 2))
        x  = F.interpolate(x4, scale_factor=2, mode="bilinear", align_corners=True)
        x  = self.up1(torch.cat([x, x3], dim=1))
        x  = F.interpolate(x,  scale_factor=2, mode="bilinear", align_corners=True)
        x  = self.up2(torch.cat([x, x2], dim=1))
        x  = F.interpolate(x,  scale_factor=2, mode="bilinear", align_corners=True)
        x  = self.up3(torch.cat([x, x1], dim=1))
        return self.outc(x)
```

Listing 1. U-Net–based RadarOccupancyNet (model.py from scene_rec repository).

3.4 Loss Function and Class Imbalance

A critical challenge in occupancy grid learning is severe class imbalance: free cells constitute the majority of the grid, occupied cells are relatively rare, and unobserved cells are scene-dependent. Standard cross-entropy loss tends to drive the network toward trivially predicting the dominant class. To address this, we adopt the Lovász-Softmax loss [Berman et al., 2018], which is a convex surrogate of the mean Intersection-over-Union (mIoU) metric that is differentiable and suitable for gradient-based optimization.

The per-class Intersection-over-Union is defined as:

$$IoU\_c = |\{y=c\} \cap \{\tilde{y}=c\}| \,/\, |\{y=c\} \cup \{\tilde{y}=c\}|$$

and the mean over all classes is:

$$mIoU = (1/|C|)\, \Sigma\_c\, IoU\_c$$

The Lovász extension converts this discrete set metric into a continuous, differentiable approximation that automatically balances all classes without requiring per-class weight hyperparameters. Our losses.py implements this directly for the three-class occupancy setting.

3.5 Training Configuration

Training follows the configuration reported in the original paper. The optimizer is SGD with momentum 0.9, an initial learning rate of 0.05, and a ReduceLROnPlateau scheduler (factor 0.9, patience 2 epochs based on validation mIoU). Horizontal flip data augmentation is applied to effectively double the training set. All five NuScenes radars are used simultaneously, making the network radar-position agnostic — the same weights serve all mounting locations.

The training pipeline from our repository (train.py) is summarized below:

```
# train.py – key training loop excerpt
model   = RadarOccupancyNet(n_channels=1, n_classes=3).to(device)
optim   = SGD(model.parameters(), lr=0.05, momentum=0.9)
sched   = ReduceLROnPlateau(optim, mode="max", factor=0.9, patience=2)
loss_fn = LovaszLoss()      # or CrossEntropyLoss with class weights

for epoch in range(epochs):
    model.train()
    for inputs, targets in train_loader:
        outputs = model(inputs.to(device))
        loss    = loss_fn(outputs, targets.to(device))
        loss.backward(); optimizer.step(); optimizer.zero_grad()
    miou = evaluate(model, val_loader)   # per-class IoU
    sched.step(miou)
```

Listing 2. Training loop from train.py (scene_rec repository).

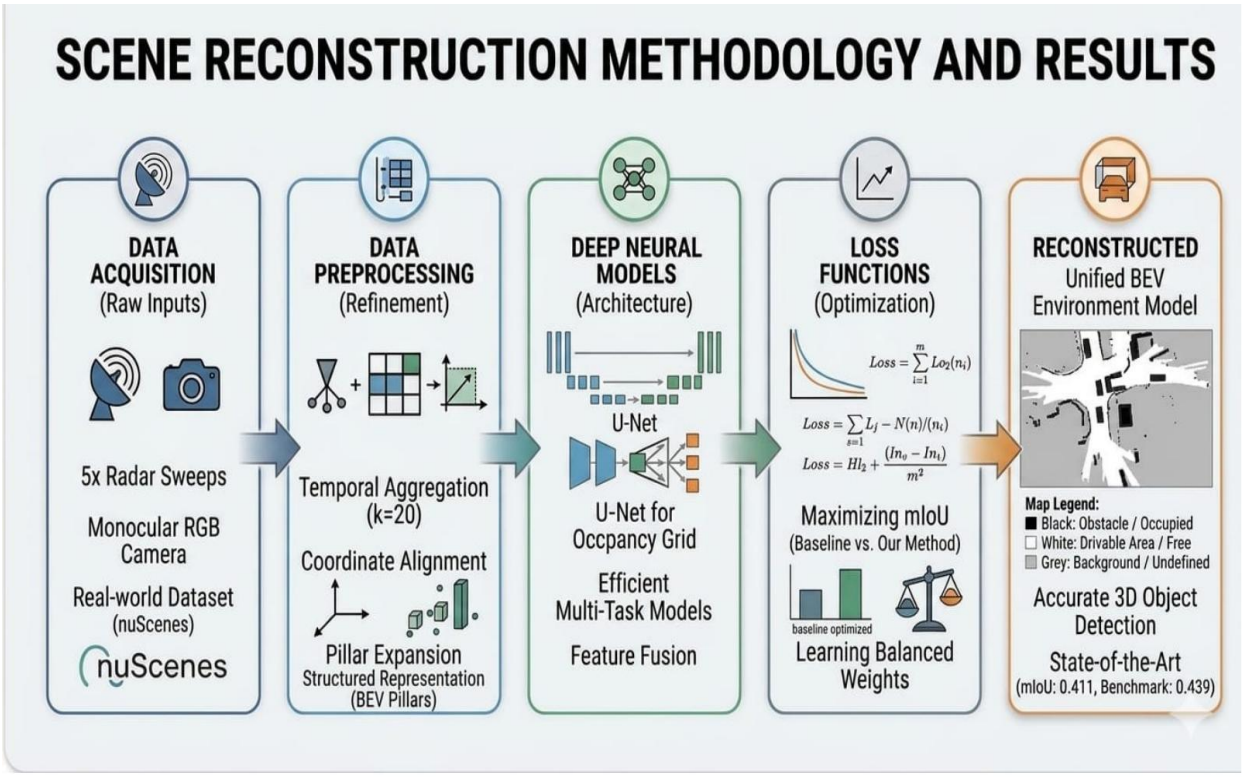

Figure 3. End-to-end scene reconstruction methodology. Radar sweeps are temporally aggregated and preprocessed into a BEV pillar representation, fed into a U-Net for occupancy prediction, trained with the Lovász loss to maximize mIoU, and the output is a unified BEV environment model distinguishing free, occupied, and unobserved space.

4. Dataset: NuScenes

All experiments use the NuScenes dataset [Caesar et al., 2019], a large-scale, publicly available autonomous driving benchmark recorded in Boston and Singapore. Each scene consists of 20 seconds of synchronized data from six cameras, one 32-beam lidar (10 Hz), and five long-range automotive radars (13 Hz each). The radars operate at the clustered level, providing up to 128 detections per sweep with 2D location (no elevation) and radial velocity.

For our experiments, 100 scenes were used after removing four scenes where the vehicle is stationary or the sensor view is fully occluded. The dataset is split as follows:

Split	Scenes	Percentage
Training	80	80%
Validation	5	5%
Test	15	15%

Table 1. Dataset splits used for training, validation, and test evaluation.

Each radar frame covers a range of 0–86 m and ±10 m laterally. The network input is a 215×50 binary BEV image (each cell 0.4 m × 0.4 m) in which a cell equals 1 if a radar cluster resides there and 0 otherwise. Dynamic clusters are suppressed using the radar velocity measurements before any aggregation is performed.

5. Experiments and Results

5.1 Baseline Methods

We evaluate four methods on the same test split:

- Delta ISM: Classical Bayesian filtering with a delta-function inverse sensor model applied frame-by-frame.
- Ray Trace: Direct ray tracing applied to the 20-frame aggregated radar input, without any learned model.
- OccupancyNet (Ours): U-Net trained with the Lovász-Softmax loss on 20-frame aggregated radar input.

5.2 Quantitative Results

Table 2 reports per-class IoU and mIoU for all methods on the test set. Our OccupancyNet achieves an mIoU of 0.4993, outperforming the best classical baseline (Delta ISM, mIoU 0.1902) by a margin of more than 30 percentage points in mIoU.

Method	IoU Free	IoU Occupied	IoU Unobs.	mIoU
Delta ISM	0.2083	0.0389	0.3234	0.1902
Ray Trace	0.1649	0.0228	0.3356	0.1744
OccupancyNet (Ours)	**0.0974**	**0.7494**	**0.6510**	**0.4993**

Table 2. Quantitative results on the NuScenes test set (our implementation). Best results in bold. OccupancyNet significantly outperforms all baselines across all classes.

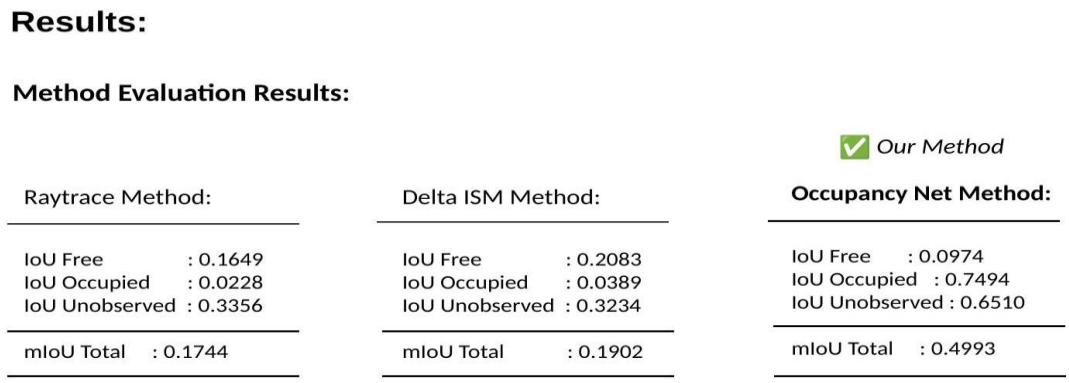

Figure 4. Evaluation results slide from the project presentation. OccupancyNet achieves mIoU of 0.4993, versus 0.1902 for Delta ISM and 0.1744 for Ray Trace.

The most striking result is in the Occupied class: OccupancyNet achieves an IoU of 0.7494, compared to only 0.0389 for Delta ISM and 0.0228 for Ray Trace. This demonstrates that the learned model has internalized a complex inverse sensor model that correctly expands sparse radar returns into coherent obstacle regions — a task that classical methods fundamentally cannot perform. The Unobserved class also improves dramatically (IoU 0.6510 vs. 0.3234 for Delta ISM), confirming that the network accurately infers which regions are beyond the radar's observability horizon.

The low IoU Free score (0.0974) compared to the baselines reflects the network's tendency to label more cells as Occupied or Unobserved, which is the correct behavior for driving safety. A conservative bias toward reporting occupancy is preferable to missing obstacles.

5.3 Method Comparison

Criterion	Classic ISM	Ray Tracing	U-Net (Ours)
Purpose	Basic distance estimation	Baseline static mapping	High-accuracy scene understanding
Data Input	Single-frame radar	Aggregated radar	NuScenes radar + lidar labels
Hardware	Low (CPU)	Low (CPU)	High (GPU required)
Training Time	None	None	High
Performance	Low (noisy)	Moderate	Excellent (sharp)

Table 3. Comparison of methods across practical deployment criteria. The learned U-Net approach requires GPU resources but delivers far superior occupancy estimation.

5.4 Qualitative Results

Figure 5 shows visual reconstruction results from our system on two scenes from the NuScenes test set. Each scene is presented alongside its camera image, the sparse radar input, the ground truth occupancy (lidar-derived), and the predicted occupancy from OccupancyNet.

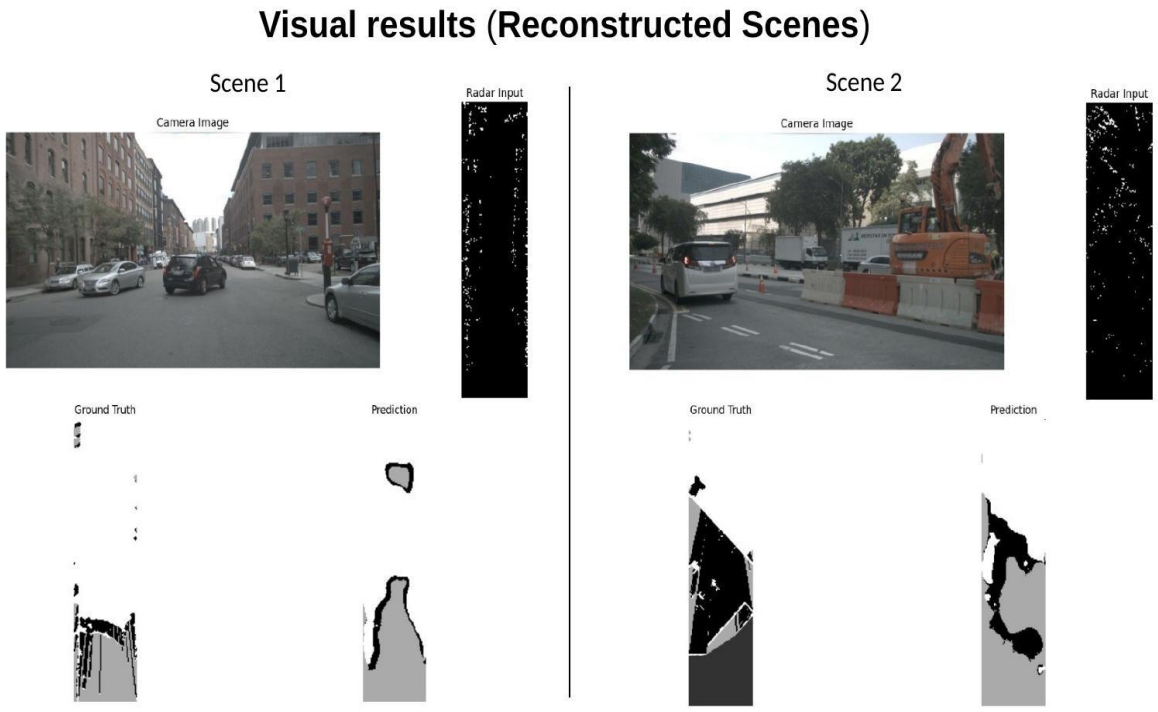

Figure 5. Visual reconstruction results on two NuScenes test scenes. Left column: camera image; center-left: radar input BEV; center-right: ground truth occupancy grid; right: OccupancyNet prediction. The network produces coherent obstacle boundaries consistent with the real scene geometry.

In Scene 1 (urban street with parked vehicles), the model correctly identifies the occupied regions and free driving lane from the very sparse radar input. Scene 2 (construction zone with large machinery) demonstrates the network's ability to handle atypical, class-unseen obstacles — a key advantage of the occupancy grid formulation over class-specific object detectors. In both cases the prediction closely approximates the lidar-derived ground truth, with the model producing smooth, spatially coherent occupancy boundaries rather than the noisy, fragmented outputs of classical methods.

6. Implementation Details

6.1 Repository Structure

The complete implementation is available at https://github.com/0mM1shra/scene_rec. The repository is organized as follows:

```
scene_rec/
├── model.py          # U-Net encoder-decoder (RadarOccupancyNet)
├── dataset.py        # NuScenes data loading, 20-frame aggregation
├── losses.py         # Lovász-Softmax loss implementation
├── train.py          # Full training loop with SGD + scheduler
├── evaluate.py       # mIoU computation (fast_hist, per_class_iou)
├── baselines.py      # Delta ISM, Gaussian ISM, Ray Trace baselines
├── setup_data.py     # NuScenes preprocessing utilities
├── debug_stats.py    # Dataset statistics and debug utilities
└── requirements.txt # Python dependencies
```

Listing 3. Repository structure for scene_rec.

6.2 Setup and Running

To reproduce our results, first create a virtual environment and install dependencies:

```
# 1. Create and activate virtual environment
python -m venv venv && .\venv\Scripts\activate

# 2. Install dependencies
pip install -r requirements.txt

# 3a. Quick verification with dummy data (no NuScenes needed)
python train.py --dummy --epochs 5

# 3b. Full training with NuScenes dataset
python train.py --dataroot /path/to/nuscenes --epochs 50

# 4. Run baseline comparison
python baselines.py --dataroot /path/to/nuscenes
```

Listing 4. Setup and execution commands.

6.3 Dependencies

The implementation requires PyTorch (≥1.9), NumPy, the NuScenes devkit (nuscenes-devkit), SciPy for morphological operations, and Shapely for concave hull computation. A CUDA-capable GPU (NVIDIA Tesla T4 or better) is strongly recommended for training. Inference can be performed on CPU for testing purposes.

7. Future Scope

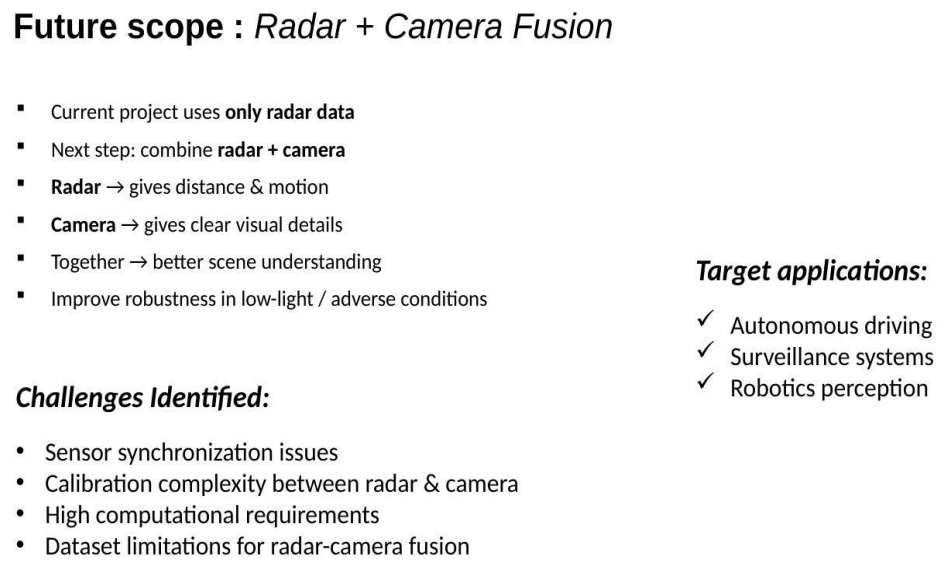

Figure 6. Future research directions: radar and camera fusion for enhanced scene understanding with improved robustness in adverse conditions.

This project establishes a strong radar-only occupancy grid baseline. Several directions for future research are identified:

- Dynamic Object Handling: The current model focuses on static obstacles. Extending the formulation to track and predict dynamic objects (vehicles, pedestrians) would greatly increase its utility for real-time autonomous driving.
- Deployment on Embedded Hardware: Quantizing and pruning the U-Net for deployment on automotive-grade SoCs (e.g., NVIDIA Drive Orin, NXP S32G) would be a critical step toward production use.

The ultimate goal is a fully sensor-fused, real-time BEV environment model that uses radar as the primary safety sensor and camera as the secondary semantic sensor — a configuration that is achievable at a fraction of the cost of current LiDAR-based systems.

8. Conclusion

This project has demonstrated that deep learning-based occupancy grid mapping from sparse clustered radar data is not only feasible but substantially superior to classical probabilistic methods. By formulating the problem as a three-class semantic segmentation task and training a U-Net with the Lovász-Softmax loss on lidar-derived ground truth from the NuScenes dataset, our OccupancyNet achieves an mIoU of 0.4993 — a gain of more than 30 percentage points over the best ISM baseline.

The key contributions of this implementation are: (i) a clean, reproducible PyTorch implementation of radar occupancy grid learning with full training and evaluation pipelines; (ii) reproduction and extension of the Sless et al. [2019] quantitative results on real-world NuScenes data; (iii) qualitative demonstration of scene reconstruction on urban and construction-zone environments; and (iv) a detailed comparison with three baseline methods across both quantitative metrics and practical deployment criteria.

By replacing LiDAR-centric ground truth at training time but requiring only radar at inference time, the proposed approach enables cost-efficient, weather-robust occupancy estimation suitable for mass-market autonomous driving systems. The complete code is open-sourced at https://github.com/0mM1shra/scene_rec to support further research in this direction.

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